RAJALAKSHMI ENGINEERING COLLEGE

Dhivya Sri G 251001055-2025-IT

D2

| | |
|---|---|
| **Started on** | Friday, 7 August 2026, 9:08 PM |
| **State** | Finished |
| **Completed on** | Friday, 7 August 2026, 9:37 PM |
| **Time taken** | 29 mins 9 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include<stdio.h>
  void Factor(int num){
      int c=0;
      for(int i=1;i<=num;++i)
     {
         c=c+1;
        if(num%i == 0)
         {
             c=c+1;
             }
             c=c+1;
         }
         c=c+1;
         printf("%d",c);
     }
     int main(){
         int num;
         scanf("%d",&num);
         Factor(num);
         return 0;
     }
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 31 | 31 | ✔ |
| ✔ | 25 | 54 | 54 | ✔ |
| ✔ | 4 | 12 | 12 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course